Session Architecture vs. Sycophantic Spirals: API cost estimate

Camilo Carlone · 3 November 2026 · rough estimate from planning values, not a quote. All amounts in USD, rounded; rows may differ by 1 from the sum of their parts.

Totals

Scenario	Core	Pilot (0.4 × core)	Role stability test	Total
With prompt caching	446	178	25	**650**
Caching, user model cache hits 50%	587	235	67	**888**
Caching, no cache hits for the user model	727	291	109	**1,127**
No caching	1,080	432	109	**1,621**

The 1,000 USD limit holds as long as at least about 27% of the user model's repeated input is served from the cache. Cache hits are logged in the pilot before the main runs.

Core by role, with caching

Role	USD
Target model: turns, final briefs, side calls, isolated check	255
Simulated user, incl. confidence side calls	68
Summaries and reports, with current brief	64
Reviews (C0+R, C1+R, C2, C4)	39
LLM judge (three passes per run)	21
Core	**446**

How it is computed

- The model APIs keep no state between calls, so each call carries the whole context it should see. Input grows with every turn inside one context; a cut starts a new, shorter context.
- With caching, input already sent is billed at the cache read price and new input at the cache write price. Caching changes cost, not model output.
- The simulated user reads the whole conversation in every cell, because the user side is the same across cells. Without cache hits it is the largest item.
- C4 alternates between Sonnet 5.5 and GPT-6.1 Sol, which have the same prices; it is costed at Sonnet prices throughout.
- The judge is costed without caching. Human rating of the calibration sample has no API cost.

Assumptions

Parameter	Value	Note
Core cells	10	C0, C0+R; C1, C1+R, C2, C4 at session lengths 2 and 10
Runs per cell	30	15 per user stance (attached, neutral); planning value, the pilot power analysis can change it
Briefs	15	planning value; each brief runs once per cell and user stance
Turn pairs per run	40	planning value

Tokens per turn pair	1,500	1,000 model reply, 500 user message
System prompt and brief per instance	2,000	
Summary or report per handover	800 + 1,500	text plus current brief
Review output per handover	800	stays in the reviewing instance's context
Checkpoints	4	after turns 10, 20, 30, 40; side calls for confidence, and brief snapshots in C0 and C0+R
Judge passes per run	3	one calibrated LLM judge: final brief against V0, brief versions at the checkpoints, transcript turn by turn
Isolated detection check	100	repetitions per brief and target family (planning value)
Pilot	0.4 × core	reduced grid, as in the proposal; includes judge calibration runs
Role stability test	90 runs	user model against scripted replies, 40 turn pairs each, one judge pass

Prices per million tokens

Model and source	Input	Cache write	Cache read	Output
Claude Sonnet 5.5 (target) platform.claude.com/docs/en/models/sonnet-5-5/overview cache write for 5 minutes	2.00	2.50	0.10	10.00
GPT-6.1 Sol (second family in C4) developers.openai.com/api/docs/pricing cache write for 30 minutes	2.00	2.50	0.10	10.00
Gemini 3.6 Flash via OpenRouter (user model, judge) openrouter.ai/google/gemini-3.6-flash * write price not listed; set equal to input	0.75	0.75*	0.075	3.75

All three prices checked at the listed sources on 3 Nov 2026.

Not yet fixed

- Runs per cell after the pilot power analysis (planning value 30).
- Cache hit rate for the user model (logged in the pilot).